

SYED TALAL JILANI

AI Engineer | Generative AI | Agentic AI | Computer Vision | Full Stack Development

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Professional Summary

AI Engineer and Full Stack Developer with professional experience building AI-powered applications, agentic workflows, business automation systems, computer vision solutions, mobile applications, blockchain systems, and IoT solutions. Hands-on experience with Generative AI, Agentic AI, LLM applications, RAG, LangChain, LangGraph, Ollama, n8n, YOLO, RT-DETR, RF-DETR, SAP HANA agents, WhatsApp automation, React Native, Node.js, Laravel, PostgreSQL, Docker, Linux, ESP32, and blockchain technologies.

Research experience in remote sensing, satellite image analysis, agricultural computer vision, and object detection benchmarking, including remote collaboration with Dr. Aslam through a research lab at Aberystwyth University, UK.

Core Skills

AI / ML & Computer Vision: Python, PyTorch, YOLO, YOLO26, RT-DETR, RF-DETR, Object Detection, Deep Learning, Model Fine-tuning, Model Benchmarking, Edge AI

Generative AI & Agentic AI: LangChain, LangGraph, Ollama, RAG, AI Agents, SQL Agents, Prompt Engineering, LLM Applications, Multi-Agent Workflows

Automation: n8n, WhatsApp Business API, WeCom, Conversational AI, Business Process Automation, Agentic Workflows

Frontend & Mobile: React, React Native, Next.js, TypeScript, JavaScript, Tailwind CSS, HTML, CSS

Backend: Node.js, Express.js, Laravel, PHP, REST APIs, Prisma

Databases: PostgreSQL, MySQL, MongoDB, Redis, SAP HANA

IoT & Embedded: ESP32, ESP32-S3, NRF24L01, KiCad, EasyEDA, SHT31, MQ-2, Capacitive Soil Moisture Sensors, Relay Control

Blockchain: Solidity, ethers.js, Web3.js, OpenZeppelin, Smart Contracts

Tools & Infrastructure: Linux, Git, GitHub, Docker, PM2, AWS, Whisper, Edge TTS, Tableau

Remote Sensing / GIS: Google Earth Engine, QGIS, ArcGIS Pro, Landsat, Sentinel-2, Satellite Image Analysis, NDVI

Professional Experience

Senior AI Engineer (progressed from AI Engineer)

Nov 2025 – Present

Evyol Group of Industries

- Designed and developed AI-powered business automation solutions across sales, customer support, supplier communication, analytics, agriculture, and industrial operations.
- Built an **ECO Pest Sales Bot** for WhatsApp using **n8n** to automate customer interactions and sales workflows.
- Re-architected the ECO Pest solution as a web-based AI application using **LangChain and Ollama** following evolving business requirements.
- Designed agentic workflows and multi-agent orchestration using **LangGraph** for business automation applications.
- Developed an **SQL Agent for SAP HANA** enabling business users to query enterprise data using natural language.
- Developed a **Tableau Bot** for conversational interaction with business reporting and analytics workflows.
- Built a **WeCom Supplier Negotiation Bot** supporting supplier communication and negotiation workflows.
- Developed a **Bloom Coffee WhatsApp Bot** supporting order booking, tracking, cancellation, and automated order-status updates.
- Developed **Zahdarnama**, an Urdu content-to-audio platform using Edge TTS and Whisper with synchronized subtitle playback.
- Fine-tuned **YOLO models** on custom number-plate datasets and developed an NPR system for factory and residential sites.
- Built an **Agricultural Data Bank React Native mobile application** for collecting crop disease and weed observations directly from agricultural fields.
- Designed role-based workflows for **Uploader, Doctor, and Administrator** users within the Agricultural Data Bank.
- Conducted computer vision research on rice and cotton disease detection using **YOLO26-L, RF-DETR-L, and RT-DETR-L**.
- Benchmarked object detection models based on detection performance and inference speed to evaluate practical deployment trade-offs.
- Designed a custom weather-monitoring PCB using **KiCad, ESP32-S3, SHT31, and MQ-2** sensors.
- Built an IoT-based soil-moisture monitoring and irrigation control system using **ESP32, NRF24, capacitive sensors, and relay control**.

Full Stack Developer

Jan 2025 – Oct 2025

AAM Group of Companies

- Developed backend functionality for the **AAM Click** platform using Laravel.
- Implemented administration and user-management functionality.
- Developed backend APIs and database functionality for application features and frontend integration.
- Designed and implemented a blockchain-based reward system using **Node.js and ethers.js**.
- Developed token reward logic based on user engagement, including commenting and post likes.
- Identified a critical vulnerability in an existing token smart contract that had been exploited.
- Redesigned and redeployed the token contract using **OpenZeppelin standards** while preserving the existing external interface to avoid breaking integrations.

Intern → Junior Developer

Jun 2024 – Dec 2024

Faisal Movers

- Maintained and enhanced the company's **Human Resource Management System (HRMS)**.
- Implemented maternity, sick, and annual leave management according to company policies.
- Developed roster and attendance logic ensuring Sundays before an employee's joining date were excluded from leave calculations.
- Developed an **Access Manager** supporting granular page-level, form-level, and report-level permissions.
- Fixed existing HRMS issues and extended modules without disrupting existing workflows.

Software Development Intern

Mar 2024 – Jun 2024

Instacom

- Migrated an existing **Radius Management System** backend from PHP to Node.js.
- Developed backend services using **Node.js, Prisma ORM, and PostgreSQL**.
- Worked in a Linux-based development and deployment environment.
- Refactored existing functionality into a more maintainable backend architecture.

Research Experience

Remote Sensing Research Collaboration

Research Collaboration

Aberystwyth University, UK – Remote

- Worked remotely with **Dr. Aslam** on **remote sensing and satellite image analysis** research through his research lab at Aberystwyth University.
- Worked with satellite imagery and remote-sensing workflows for environmental and agricultural analysis.
- Explored satellite-derived vegetation and environmental indices including **NDVI**.
- Worked with geospatial datasets and image-processing workflows for extracting information from satellite imagery.
- Contributed to research-oriented experimentation, analysis, and technical development in a remote international research collaboration.

Computer Vision Researcher

2025 – Present

Plant Disease Detection Benchmark

- Conducted comparative research on modern object detection models for rice and cotton plant disease detection.
- Evaluated **YOLO26-L, RF-DETR-L, and RT-DETR-L** based on detection performance and inference speed.
- Investigated accuracy-versus-computational-efficiency trade-offs for agricultural AI applications.
- Prepared a research manuscript based on experimental results.

Selected AI Projects

ECO Pest AI Sales Assistant

LangChain, Ollama, n8n, WhatsApp

- Built an AI-powered sales assistant for automating customer interactions and sales workflows.
- Initially implemented WhatsApp automation using n8n and later developed a web-based LangChain + Ollama version.

SAP HANA Natural Language SQL Agent

LangChain, Ollama, SAP HANA, SQL

- Developed an AI agent allowing business users to query SAP HANA databases using natural language.
- Designed the workflow to interpret questions, generate SQL, execute queries, and return business information.

Zahdarnama – Urdu Audio Narration Platform

Python, Edge TTS, Whisper

- Built an Urdu content-to-audio platform for converting magazine articles into narrated audio.
- Integrated Edge TTS for Urdu speech generation and Whisper for speech-to-text processing.
- Implemented synchronized subtitle highlighting based on the spoken audio timeline.

Bloom Coffee WhatsApp Ordering Bot

WhatsApp, AI Automation, Backend APIs

- Developed a WhatsApp-based ordering assistant supporting order booking, tracking, cancellation, and status updates.

WeCom Supplier Negotiation Bot

WeCom, AI Agents, Automation

- Developed an AI-powered bot supporting supplier communication and negotiation workflows.

Computer Vision Projects

AI Number Plate Recognition System

Python, YOLO, Computer Vision

- Fine-tuned YOLO models on custom number-plate datasets for automated vehicle detection.

- Developed the system for factory and residential security environments.

Plant Disease Detection Benchmark

YOLO26-L, RF-DETR-L, RT-DETR-L

- Benchmarked modern object detection architectures using rice and cotton disease datasets.
- Compared detection performance and inference speed across YOLO26-L, RF-DETR-L, and RT-DETR-L.

Agricultural Data Bank – React Native Mobile App

React Native, Agriculture, Database

- Developed a **React Native mobile application** for collecting crop disease and weed observations directly from agricultural fields.
- Designed role-based workflows for **Uploader, Doctor, and Administrator** users.
- Created a mobile-first field-data collection workflow for agricultural observations.
- Structured collected data to support centralized management and future AI/computer vision model training.

IoT & Embedded Projects

Soil Moisture Monitoring & Automated Irrigation

ESP32, NRF24, Sensors, Relay

- Built an IoT-based soil moisture monitoring system using capacitive soil sensors and ESP32.
- Implemented NRF24 wireless communication between sensor and control nodes.
- Developed relay-based control for automated water pump and tubewell operation.

ESP32-S3 Weather Monitoring Device

ESP32-S3, KiCad, SHT31, MQ-2

- Designed a custom PCB in KiCad around an ESP32-S3.
- Integrated SHT31 temperature/humidity and MQ-2 gas sensors.
- Designed the hardware architecture for environmental monitoring and IoT connectivity.

University & Academic Projects

GatherNow – Google Meet Clone

WebRTC, Node.js, React, Socket.io

- Developed a real-time video conferencing platform inspired by Google Meet as the **Bachelor's Final Year Project**.
- Implemented real-time video and audio communication using **WebRTC**.
- Developed real-time signaling and communication workflows using **Node.js and Socket.io**.
- Built the frontend interface for meeting creation, joining, and real-time communication.
- Worked on peer-to-peer communication and real-time application architecture.

Me AI Assistant – Voice-Based AI Assistant

AI, Voice, Chrome Extension

- Built a voice-enabled AI assistant as a Chrome extension for interacting with AI through natural language.
- Integrated AI APIs and voice interaction into a browser-based assistant workflow.
- Received a **Best AI/ML API Usage** award during project evaluation.

QuickError / LogTracker

Node.js, NPM

- Developed an NPM package for simplifying application error and log tracking in Node.js applications.

WithEasiness AI Tools

Next.js, AI, Full Stack

- Developed AI-powered web utilities using Next.js.
- Built tools including AI video-generation workflows and an AI-powered job-description validation system.

University Activities & Leadership

Section Leader

2025

Stanford Code in Place

- Selected as a Section Leader for Stanford Code in Place 2025.
- Mentored students learning programming and core computer science concepts.
- Conducted programming sessions and provided technical guidance on exercises and projects.

Head – Open Source Development Club

University Leadership

NFC Institute of Engineering & Technology

- Led university open-source development activities and student-focused software projects.
- Guided students in practical software engineering, collaborative development, and open-source contribution.
- Encouraged students to build practical applications and participate in collaborative technical development.

Research Publication

A Unified Cross-Family Benchmark of Object Detection Models for Plant Disease Detection

- Research manuscript comparing YOLO26-L, RF-DETR-L, and RT-DETR-L for rice and cotton plant disease detection.
- Focused on comparative detection performance and inference efficiency.
- **Status:** Submitted for journal publication.

Technical Skills

Programming: Python, JavaScript, TypeScript, PHP, Solidity, SQL, C/C++

AI / Machine Learning: PyTorch, YOLO, YOLO26, RT-DETR, RF-DETR, Computer Vision, Object Detection, Deep Learning, NLP, LLMs, Generative AI, Model Fine-tuning, Model Benchmarking

Generative AI / Agentic AI: LangChain, LangGraph, Ollama, RAG, AI Agents, SQL Agents, Prompt Engineering, n8n, Multi-Agent Workflows

Backend: Node.js, Express.js, Laravel, REST APIs, Prisma, PHP

Frontend / Mobile: React, React Native, Next.js, TypeScript, JavaScript, Tailwind CSS, HTML, CSS

Real-Time Systems: WebRTC, Socket.io, Peer-to-Peer Communication, Real-Time Applications

Databases: PostgreSQL, MySQL, MongoDB, Redis, SAP HANA

Blockchain: Solidity, ethers.js, Web3.js, OpenZeppelin, Smart Contracts

IoT / Embedded: ESP32, ESP32-S3, NRF24, SHT31, MQ-2, Capacitive Soil Moisture Sensors, Relay Control, PCB Design, KiCad, EasyEDA

DevOps / Systems: Linux, Git, GitHub, Docker, PM2, AWS

GIS / Remote Sensing: Google Earth Engine, QGIS, ArcGIS Pro, Landsat, Sentinel-2, Satellite Image Analysis, NDVI

Education

Bachelor of Science in Computer Science (BSCS) 2020 – 2024
NFC Institute of Engineering & Technology, Multan, Pakistan

Final Year Project: *GatherNow – Google Meet Clone*

Relevant Areas: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Database Systems, Software Engineering, Web Development, Computer Networks, Remote Sensing, Geographic Information Systems

Professional Interests

Agentic AI | Generative AI | AI Automation | Enterprise AI | Computer Vision | Multimodal AI | Remote Sensing | Agricultural AI | Intelligent IoT Systems